Bipartite networks of cancer genes associated with specific drugs using pharmacogenomics

Alberto Berral-González[1], Mónica M. Arroyo[1,2], Diego Alonso-López[1], Santiago Bueno-Fortes[1], Javier De Las Rivas[1]

[1] Bioinformatics and Functional Genomics Group, Cancer Research Center (CiC-IMBCC, CSIC/USAL/IBSAL), Consejo Superior de Investigaciones Científicas (CSIC) and University of Salamanca (USAL), 37007 Salamanca, Spain.
[2] Department of Chemistry, Pontifical Catholic University of Puerto Rico (PCUPR), Ponce, Puerto Rico.

Corresponding Author: Alberto Berral-González (aberralgonzalez@usal.es)
Co-corresponding Author: Monica M Arroyo (monica_arroyo@pucpr.edu)

Summary

The study of the genome-wide activity of the genes (i.e. the expression of the genes or their genetic variation) may be related to differences in the effects of drugs. Pharmacogenomics allows better mapping of the targets of cancer drugs and potential interacting secondary agents, but there are many drugs whose mechanisms of action have not been fully deciphered [1]. Complex diseases such as cancer, with hundreds of clinically approved drugs could benefit from integrating these data in clinical trials and drug development [2, 3]. The study of these drug-targets to better understand their mechanism of action could lead to possible new treatments or an improvement of existing ones by overcoming drug resistance [4]. This study comprised a large-scale screening analysis to find new associations between chemical substances (non-FDA-approved cancer drug) and human genes using transcriptomic profiling [6]. The results showed 327 genes and 2733 drugs have 6183 positive correlations, across the sixty cancer cell lines (using a threshold of Pearson coefficient r < 0.60 and p-value ≤ 0.05).

Methods

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We began with 705 genes (oncogenes, fusion, and tumour suppressor genes (TSG)) CellMiner [6] and COSMIC cancer gene census (CGC) [7] to calculate the Pearson and Spearman correlations with 21,738 drugs. Correlations with a coefficient value ≥ 0.334 (p ≤ 0.05) were chosen to create several subsets for further analysis. The results were filtered to generate a set with molecules that includes drugs approved for non-cancer therapies or drug candidates [6]. Next, global bipartite networks were built to further study the interactions between compounds and their targets, to better unravel their mechanisms of action. In addition, these data were included in GEDA, an online web-tool that allows the user to navigate the networks obtained. Furthermore, the user can access the data used and review the results obtained in this and in our previous work.

Highlights

- The purpose of this study is to explore and discover new "drug-target candidates" in cancer.
- Our results show new plausible cellular targets based on gene expression and drug activity.
- This study may contribute to further study non-FDA-approved cancer drugs as viable therapeutics for drug resistant cancers.

Data integration

- Web App
- Interactive
- Browsable data
- Visual results

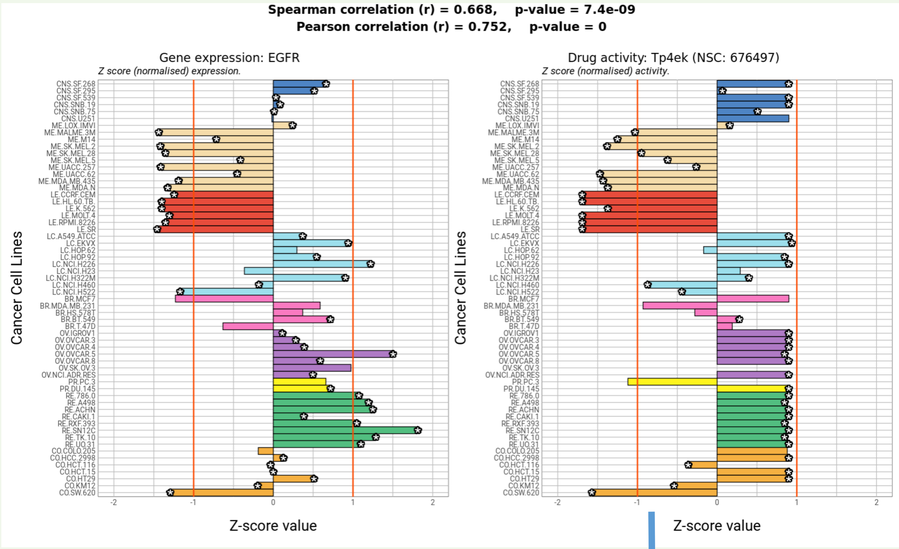
GEDA APP LINK

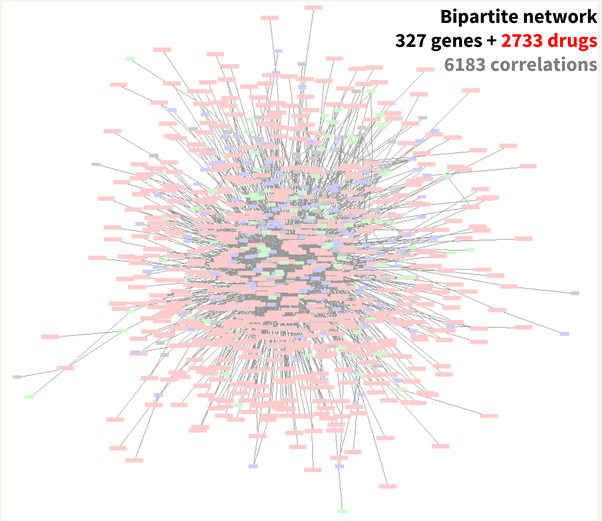
Results

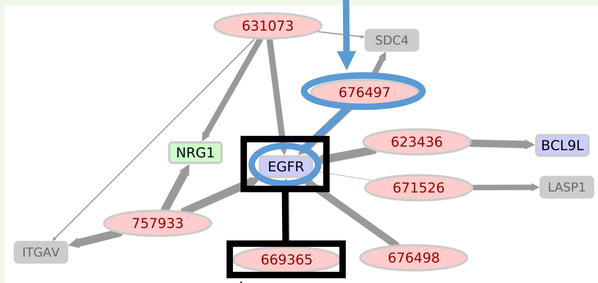
Left panel: **EGFR** gene expression.

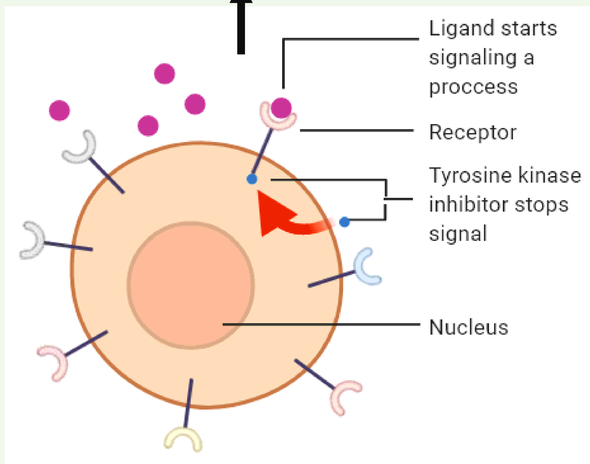
Right panel: **Tp4ek** drug activity. **Tp4ek** is a recombinant exotoxin that interacts with **EGFR.**

Star indicates that the expression and activity levels are in the same direction for cell line. Orange vertical line indicates -1 ≤ Z-score ≥ 1.









Top left panel: Global bipartite network for the 6183 Gene – Drug correlations. **Drugs** are represented in **red ellipses**. **Genes** are represented in colors according to their type: Fusion, Oncogenes, and TSG (grey, blue and green, respectively)

Top right panel: **EGFR –669365** interactions subnetwork. In **blue** is marked **EGFR – Tp4ek** interaction. **EGFR – NSC 669365** interaction is marked in **black**.

Bottom right panel: **NSC 669365** drug mechanism of action. This drug is a tyrosine kinase inhibitor.

Our results, with supporting evidence from the literature, give credence to our proposal that some of these drug candidates may serve as novel therapeutics, and that other small molecules should be further characterized. We are currently performing an exhaustive analysis of potential drug candidates based on our initial results.

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